



OLLSCOIL NA GAILLIMH
UNIVERSITY OF GALWAY

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CT5127 Computer Networks & Data Communications

Week 1: Module Overview

Learning Outcomes (LOs)

- Define the theory, concepts and principals of data communications
- Define the OSI model and describe the layers that computer systems use to communicate over network
- Explain the way protocols currently in use in the internet work
- Capture and analyse network traffic
- Use scripting tools to automate networking tasks



Module Organisation

- Breakdown of lectures:
 - Part 01: Week 1 – 6: Covered by me
 - Part 02: Week 7 – 12: Will be covered by Dr Xue Yang



Indicative Content

- **Part 01:**

- Introduction to Computer Networks
- Network Architectures (Reference Models) and Network Devices
- Physical layer: Modulation
- Link Layer: Frames + ARP
- Link Layer: Error Control
- Link Layer: Flow Control

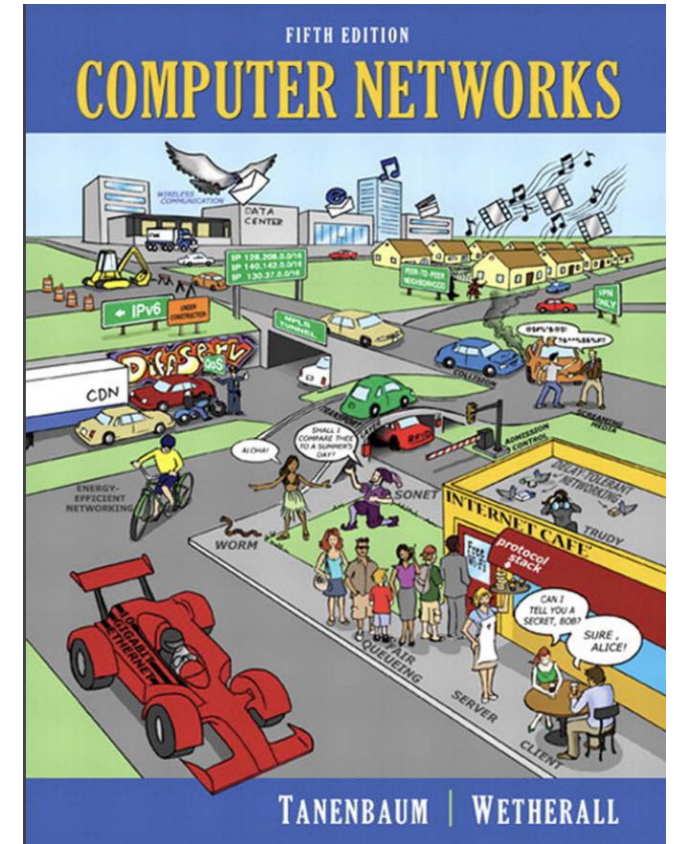
- **Part 02:**

- Network Layer
- Transport Layer
- Application layer
- Network Utilities



Course Description

- Based on sections of the book
 - Computer Networks, Global Edition
by [Andrew Tanenbaum](#), [Nick Feamster](#), [David Wetherall](#)



- ISBN-10 **1292374063**
- ISBN-13 **978-1292374062**
- Edition **6th**
- Publisher **Pearson**
- Publication date **3 Mar. 2021**

Lectures and Labs

- **Lecture:**

- Tuesdays 11-13:00
- From Week 1 to 6 (with me)
- Lectures will be posted every week through Canvas
 - Slides of the lecture
 - Additional content (if any):

- **Labs**

- Thursdays 10-12:00
- From Week 1 to 12 with the teaching assistant (TA)



Assessments

Overall module:

- 20% Continuous Assessment
- 80% Final Written Exam

• Part 1:

- 10% Continuous Assessment
 - **5% Quizzes:**
 - You will have one quiz each week.
 - Each quiz is worth 1%
 - The quiz will be about the learning material of that week
 - You will have **one week** to complete the quiz (directly on Canvas)
 - You will be able to have **three** attempts within the week (best score is kept)
 - **5 % test:**
 - In class test
 - Details to come later



Lecture Related Questions

- If you ever have any questions, whether they're related to the course content, assignments, or anything else, please don't hesitate to reach out to me.
- The best way to contact me is via email:
malika.bendechache@universityofgalway.ie
- I will endeavour to respond within 48 hours.
 - If you would like to send me an email, use your University of Galway email account and state in the email: name, Module code CT5127, and Student ID.

